

ASSIGNMENT

BUSINESS ANALYTICS TERMINOLOGY.

FOUNDATIONAL CONCEPTS OF ANALYTICS.

1. BUSINESS ANALYTICS.

Is the use of data and analysis to understand how a business is performing and to help make better decisions instead of relying on guesswork.

For Example

Imagine a clothing store notices that more people buy jackets when it gets colder. By looking at their past sales data, they can prepare by stocking more jackets before winter starts and even run special promotions to increase sales.

Source

IBM

2. DATA ANALYTICS.

Is the process of examining raw data to find patterns and useful insights that help businesses make better decisions.

For Example.

A supermarket studies buying habits and notices that customers often buy bread and milk together, so they place these items close to each other to improve sales.

Source

IBM.

3) BUSINESS MET

3) BUSINESS INTELLIGENCE

Is the process of using data tools and reports to understand how a business is performing and support decision making.

Example

A company uses sales dashboards to track performance and notices a drop in one region, then takes action to improve it.

Source

IBM.

4) DATA-DRIVEN MAKING.

Is when decisions are made using facts and data instead of assumptions or personal opinions.

Example

A restaurant uses sales data and customer feedback to see that chicken meals are most popular, so they add more chicken dishes to the menu. Source (IBM)

5) BIG DATA.

Is a very large and complex sets of data that cannot be easily processed using traditional methods.

Example.

Social media platforms analyze huge amounts of user activity like posts, likes and comments to understand behaviour and improve content.

Source

IBM.

6) DESCRIPTIVE ANALYTICS.

Is the process of analyzing historical data to understand what has already happened in a business. It focuses on summarizing past information using reports, charts and statistics.

Example.

A company looks at last month's sales report to see total sales, best-selling products and top-performing regions.

Source

IBM

7) DIAGNOSTIC ANALYTICS.

Is the process of analyzing data to understand why something happened. It goes beyond describing what happened and focuses on finding the cause of a problem or trend.

Example.

A company sees a drop in sales and find out it was caused by a competitor running a discount campaign.

Source

IBM

8) PREDICTIVE ANALYTICS.

Is the use of historical data, statistical models and machine learning to predict what is likely to happen in the future. It helps businesses make forecasts based on patterns and trends.

Example .

A mobile network predicts which customer may cancel their contracts and offers them discounts to keep them.

Source

IBM.

9) PRESCRIPTIVE ANALYTICS.

Is the use of data algorithms and models to recommend the best possible actions a business should take to achieve a desired outcome. It goes beyond predicting what will happen and suggests what should be done.

Example .

A delivery company uses data to recommend the best routes for drivers to save time and fuel.

Source

IBM

10) DATA LITERACY

Is the ability to read, understand, analyze and communicate with data. It means being able to make sense of data and use it to make sense of data and use it to make informed decisions in everyday work or life.

Example .

An HR officer analyzes turnover data and improves onboarding after noticing most employees resign within the first three months

Source

IBM.

DATA, MEASUREMENT AND REPORTING.

11) STRUCTURED DATA.

Is data that is organized in a fixed format, usually in rows and columns, making it easy to store, search and analyze in databases or spreadsheets.

Example.

A company's employee database showing name, id numbers, salaries and departments arranged neatly in a table.

Source.

IBM.

12) UNSTRUCTURED DATA.

Is data that does not have a fixed format or structure, making it harder to organize and analyze using traditional tools like spreadsheets or databases. It often includes text, images, videos and social media posts.

Example.

Customer reviews on social media or emails sent to a company that contain free-form text, opinions and no set format.

Source

IBM.

13) DATA WAREHOUSE.

Is a central system where a business stores large amounts of data from different sources in one place so it can be used for reporting, analysis and decision-making.

Example

A retail company stores sales and customer data from all stores and online platforms in one system to analyze performance.

Source .

IBM .

14) DATA MINING.

Is the process of examining large sets of data to discover patterns, relationships and useful insights that are not immediately obvious.

Example

A supermarket finds that customers who buy diapers often also buy baby wipes, so they place them close together in the store.

Source .

IBM .

15) DATA VISUALIZATION.

Is the process of presenting data in visual formats like charts, graphs, maps and dashboards so that information is easier to understand and interpret .

Example .

A company uses a bar chart to compare monthly sales and easily see which months performed best and worst.

Source

IBM .

16) DASHBOARD

Is a visual tool that displays key data and performance indicators in one place, usually using charts, graphs and tables. It helps users quickly monitor and understand business performance.

Example.

A sales manager uses a dashboard to track daily sales, number of customers and top-selling products in real time to make quick decisions.

Sources.

IBM.

17) METRIC

Is a measurable value used to track and assess the performance of a business, process or activity over time.

Example.

A company uses 'monthly sales revenue' as a metric to measure how much money is made each month and track performance growth.

Source.

IBM.

18) KEY PERFORMANCE INDICATOR (KPI).

Are specific measurable values used to evaluate how well a business or individual is achieving important goals.

Example.

A customer service team uses average response time to customer

queries as a KPI to measure how quickly they respond to customers.

Source .

IBM .

19) BENCHMARK.

Is a standard or point of reference used to compare performance, processes or results against others or against best practices.

Example .

A company compares its customer service response time to the industry average to see if it is performing better or worse than competitors.

Source

IBM .

20) SEGMENTATION .

Is dividing data or customers into smaller groups based on shared characteristics to make targeting and analysis easier.

Example .

A clothing store groups customers by age and buying habits to send more relevant promotions.

21) Source .

IBM .

CUSTOMER AND SALES KPIs.

21) CONVERSION RATE

Is the percentage of people who take a desired action out of the total number of visitors.

Example

If 1 000 people visit a website and 50 make a purchase the conversion rate is 5%

Source.

HubSpot.

22) CUSTOMER ACQUISITION COST (CAC)

Is the amount a business spends to gain one new customer.

Example.

If a company spends £10 000 to get 100 customers, the cost per customer is £100.

Source -

Hubspot.

23) CUSTOMER LIFETIME VALUE (CLV/LTV)

Is the total amount of money a business expects to earn from a customer over the entire time they remain a customer.

Example.

If a customer spends £500 per month for 12 months, their total value is £6 000.

Source.

Investopedia.

24) CHURN RATE.

Is the percentage of customers who stop using a company's product or service over a certain period of time.

Example.

If 200 customers start and 20 leave, the churn rate is 10%.

Source.

Investopedia.

25) CUSTOMER RETENTION RATE.

Is the percentage of customers a business keeps over a specific period of time. It shows how well a company maintains its existing customers instead of losing them.

Example

A company started with 100 customers and ended with 110 after gaining 20 new ones. After excluding the new customers, it retained 90 of its original customers, resulting in a 90% retention rate.

2 Source

Investopedia.

26) NET PROMOTER SCORE (NPS)

Is a metric used to measure customer loyalty and satisfaction. It shows how likely customers are to recommend a company's product

or service to others.

Example.

Out of 100 customers surveyed, 60 were promoters and 15 were detractors. Subtracting detractors from promoters gives an NPS of 45, showing strong customer loyalty.

Source.

Qualtrics.

27) AVERAGE ORDER VALUE (AOV)

Is the average amount of money customers spend each time they place an order. It helps businesses understand how much revenue they earn per transaction.

Example.

A store earned £5 000 from 100 orders, meaning each order was worth £50 on average.

Source

Shopify.

28) SALES FUNNEL.

Is a model that shows the journey potential customers go through from first becoming aware of a product or service making a purchase. Its called a 'funnel' because many people enter at the top but fewer reach the final stage.

Example.

Out of 1 000 people reached, only 40 ended up making a purchase showing how customers decrease at each stage of the sales funnel.

Source
Hubspot.

28) MARKETING AND CAMPAIGN ANALYTICS.

29) MARKETING CAMPAIGN.

Is a planned set of coordinated activities (such as ads, emails, social media posts or promotions) designed to promote a product or service and achieve a specific goal, like increasing sales or brand awareness.

Example.

A clothing brand ran a summer sale using social media ads, emails and discounts to promote 30% off items and increase seasonal sales.

Source
Hubspot.

30) POST-CAMPAIGN ANALYSIS.

Is the process of reviewing and evaluating a marketing campaign after it has ended to measure its performance, determine whether goals were achieved and identify what worked well or needs improvement.

Example.

After a social media campaign, the company found that website traffic increased by 40% but few visitors made purchases, so they decided to improve their landing page for better results.

Source

Google Analytics.

31) INCREMENTAL REVENUE.

Is the additional income a business earns as a direct result of a specific action, such as marketing campaign, promotion or product change, compared to what it would have earned without that action.

Example.

A company's sales increased from £50,000 to £65,000 after a campaign, resulting in £15,000 extra (incremental revenue).

Source

Investopedia.

32) RETURN ON INVESTMENT (ROI)

Is a financial metric used to measure the profitability of an investment. It shows how much profit or loss is made compared to the cost of the investments.

Example.

A company spends / spent £2 000 on a campaign and earned £3 000 in sales, making a profit of £1 000 which equals a 50% return on investment.

Source.

Investopedia.

33) RETURN ON AD SPEND (ROAS)

Is a marketing metric that measures how much revenue a business earns for every amount spent on advertising. It shows how effective an

advertising campaign is at generating sales.

Example.

A company spent £1000 on ads and earned £4000 in revenue meaning it made £4 for every £1 spent on advertising (ROAS of 4).

Source.

Shopify.

34) Click-Through Rate (CTR)

Is a digital marketing metric that measures the percentage of people who click on an advertisement, link, or email after seeing it. It shows how effective the content is at attracting attention and encouraging action.

Example.

An ad was shown 1000 times and received 50 clicks, resulting in a 5% click-through rate (CTR).

Source.

Google Ads.

35) BOUNCE RATE.

Is the percentage of visitors who visit a website and leave without clicking on anything, visiting another page or taking any action.

Example

Out of 200 website visitors, 80 left without interacting, resulting in a 40% bounce rate.

Source

Google Analytics.

36) A/B TESTING.

Is a method of comparing two versions of something (such as webpage, email or advertisement) to see which one performs better based on a specific goal, like clicks, sales or engagement.

Example.

A company tested two email subject lines and found that Version A (Get 20% off Today) received more clicks than Version B, making it the better-performing option.

Source.

Optimizely.

37) LEAD GENERATION.

Is the process of attracting and capturing potential customers (called leads) who have shown interest in a company's product or service with the goal of turning them into paying customers.

Example.

A real estate company offered a free guide and people who downloaded it shared their details becoming leads for future marketing.

Source

Hubspot.

28) MARKETING ATTRIBUTION.

Is the process of identifying and assigning credit to the marketing channels or touchpoints that influenced a customer to make a purchase or complete a desired action.

Example .

A customer saw a product on Instagram, clicked a Google ad and later bought it after an email - marketing attribution is used to identify which of these channels influenced the purchase.

Source .

Google Analytics .